

JUSTIFICATION NOTES: NEO IMPACT RISK AND MIR/IR PLANETARY DEFENSE SYSTEM (KINETIC)

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1 Executive Summary and Purpose

This document consolidates the evidence base justifying Phase C of the MIR/IR telescope (DWG-MIR-01) and the future phase of kinetic interceptors of the ASTRA/L-RAIL mission. The justification rests on three pillars: (1) recent events exposing the systemic failure of early detection (the “Green Lantern” bolide 2015; Chelyabinsk 2013; Falcon 9 lunar impact, 2026); (2) the 2029 flyby of (99942) Apophis as a reminder of vulnerability, including the hypothetical scenario of lunar impact with collateral damage to satellites; and (3) the physical and logistical advantage of the Moon as a detection and response platform ($v_{esc} = 2.38$ km/s, intact geological record, PSR cryogenic environment).

2 Evidence of the Threat: Verified Events

2.1 The “Green Lantern” bolide (07-30-2015): systemic alert failure

On July 30, 2015, a bolide of estimated magnitude < -12 (several times the brightness of the Full Moon) crossed the night sky from San Luis (Argentina) to Uruguay. Its exceptionally shallow trajectory ($\sim 4^\circ$) and long duration (> 60 s) indicate a metric-sized object (estimated between 8.5 and 20 m) that was not detected by any early warning system prior to atmospheric entry.

- **Implication:** As documented, the operational warning for objects of this type occurs at 80–90 km altitude, without time to issue alerts to the population. The kinetic energy was estimated between 230 and 700 kilotons (kt), with a potential devastation radius of 4–6 km. A similar event over a densely populated area, such as the Autonomous City of Buenos Aires (radius ~ 3.5 km, $\sim 200,000$ inhabitants per commune), would have catastrophic consequences.
- **Lesson:** Statistical probability gives an average, not a date. The event demonstrates that the threat is real and unpredictable, and that current detection systems are insufficient for small objects with shallow trajectories.

2.2 Chelyabinsk (2013) and Tunguska (1908): scale of consequences

- **Chelyabinsk:** An asteroid of ~ 17 m and $\sim 10,000$ t entered the atmosphere at 18 km/s with an angle of 20° , generating a bolide of 32.5 s duration. The airburst at 14–20 km altitude released an energy of 500 kt (2.092×10^6 GJ), causing 1,200 injuries and material damage over a wide region.
- **Tunguska:** In 1908, an object of ~ 80 m in diameter exploded over Siberia with an estimated energy of 30 Mt, felling trees over an area of 26 km radius.
- **Campo del Cielo (Argentina):** The presence of multiple large fragments (37 t in 1967, 15 t in 1576, 14.8 t in 2005, 10 t in 1997) demonstrates the historical recurrence of meteoritic impacts in the region and local vulnerability.

2.3 Falcon 9 lunar impact (08-05-2026): the calibrated experiment

The second stage of a Falcon 9 rocket ($\sim 4,000$ kg) impacted the lunar surface, near the Einstein crater (~ 181 km in diameter, located on the near side), at a velocity of $v = 2.43$ km/s (final tracking value; in case of any discrepancy, 2.43 km/s is used as it is the most precise datum). This event constitutes a calibrated impact of known mass and velocity.

- **Observed results:** The Danuri lunar orbiter (South Korea) confirmed a new crater of $\sim 18 - 30$ m in diameter and ~ 3.6 m in depth, with preliminary double morphology (due to the masses concentrated at the ends of the stage) and an asymmetric ejecta ray.
- **Lesson:** This event experimentally validates impact scaling models (Holsapple, 1993) and confirms that even objects of modest mass can generate craters and disperse material over significant distances (5–50 km).

Event	Year	Size / Energy	Consequence or lesson
Green Lantern (ARG/URU)	2015	8.5–20 m / 230–700 kt	Nonexistent detection; shallow entry 4°
Chelyabinsk (RUS)	2013	17 m / 500 kt	1,200 injured; urban shock wave
Tunguska (Siberia)	1908	~ 80 m / ~ 30 Mt	Devastation of 26 km; airburst
Falcon 9 $\rightarrow$ Einstein	2026	4 t / 1.18×10^{10} J	Calibrated crater 18–30 m; validates π -scaling
Apophis (flyby)	2029	315–370 m / ~ 0.9 –1.3 Gt (if it impacted)	Crosses inside the GEO ring

3 Impact Physics with Dimensional Verification (SI)

To ensure the robustness of the model before a technical audience (JPL/Caltech), the dimensional analysis of the fundamental equations is presented.

3.1 Kinetic energy of the impactor

$$E_k = \frac{1}{2}mv^2, \quad [E_k] = \text{kg} \cdot \left(\frac{\text{m}}{\text{s}}\right)^2 = \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} = \text{J} \quad (1)$$

Falcon 9 case: $E_k = \frac{1}{2}(4,000 \text{ kg}) \cdot (2,430 \text{ m/s})^2 = 1.18 \times 10^{10} \text{ J} \approx 2.82 \text{ t of TNT}$ (1 t TNT = $4.184 \times 10^9 \text{ J}$).

3.2 Ballistic range (D) and time of flight (t) of ejecta in lunar vacuum

$$D = \frac{v_e^2 \sin(2\theta)}{g_L}, \quad [D] = \frac{(\text{m/s})^2}{\text{m/s}^2} = \text{m} \quad (2)$$

$$t = \frac{2v_e \sin \theta}{g_L}, \quad [t] = \frac{\text{m/s}}{\text{m/s}^2} = \text{s} \quad (3)$$

For $v_e = 200 \text{ m/s}$, $\theta = 30^\circ$, and $g_L = 1.62 \text{ m/s}^2$, we obtain $D \approx 21.4 \text{ km}$ and $t \approx 123.5 \text{ s}$. The units of time and acceleration cancel cleanly, leaving only meters and seconds. These results are consistent with the ejecta ray of 5–50 km observed by Danuri.

3.3 Crater scaling (π -scaling) and Froude invariant

Crater scaling in the gravitational regime (Melosh 1989; Holsapple 1993) establishes that:

$$D_t \propto \left(\frac{E}{\rho_t} \right)^{1/3.4} g^{-1/3.4} \quad (4)$$

The validity for extrapolating from a 4 t impact to asteroidal scales is guaranteed by the dimensionless invariant, or Froude Number (π_2):

$$\pi_2 = \frac{g_L \cdot L}{v^2}, \quad [\pi_2] = \frac{(\text{m/s}^2) \cdot \text{m}}{(\text{m/s})^2} = \text{dimensionless} \quad (5)$$

The fact that the Falcon 9 impact and the formation of the Einstein crater share the same physical framework in vacuum is guaranteed by these invariants. The Einstein crater is simply the same experiment at kilometer scale.

3.4 Density profile of the lunar regolith (input to the excavation model)

- **Surface (0–15 cm):** $\rho \approx 1.3 \text{ g/cm}^3$ (fluffy regolith; Carrier et al. 1991).
- **Megaregolith (> 1 m):** $\rho \approx 1.9\text{--}2.2 \text{ g/cm}^3$ (compacted material).
- **Highlands crust (GRAIL data):** $\rho \approx 2.55 \text{ g/cm}^3$ (Wieczorek et al. 2013).

The 3.6 m crater from Falcon 9 excavated and ejected compacted megaregolith with a mean density of $\sim 2.0 \text{ g/cm}^3$.

4 Hypothetical Scenario (H-2029): Apophis — Earth / Moon / Satellite Impact Calculation

(99942) Apophis is an Aten asteroid with an orbit that will bring it to a close encounter with Earth on April 13, 2029, passing at a distance less than that of geostationary satellites. This scenario, although improbable, serves as a case study for planetary defense.

4.1 Impactor parameters

- **Adopted diameter:** $D = 315 \text{ m}$ (observed range 315–370 m).
- **Mean density (Sq type):** $\rho = 2.6 \text{ g/cm}^3$ ($2,600 \text{ kg/m}^3$).
- **Mass:**

$$m = \rho \cdot \frac{\pi D^3}{6} = 2,600 \times \frac{\pi(315)^3}{6} \approx 4.3 \times 10^{10} \text{ kg} \quad (6)$$

- **Encounter velocity:** $v_\infty \approx 7.4 \text{ km/s}$ (relative to Earth, according to CNEOS/JPL).
- **Impact velocities:**
 - **Earth:** $v_{imp,\oplus} = \sqrt{v_\infty^2 + 11.19^2} \approx 13.4 \text{ km/s}$.
 - **Moon:** $v_{imp,L} = \sqrt{v_\infty^2 + 2.38^2} \approx 7.8 \text{ km/s}$.

4.2 Case A: Impact on Earth

The kinetic energy dissipated in the collision would be:

$$E_{\oplus} = \frac{1}{2}(4.3 \times 10^{10})(13,400)^2 \approx 3.9 \times 10^{18} \text{ J} \approx 920 \text{ Mt of TNT} \quad (7)$$

This would result in a complex crater of $\sim 6\text{--}9$ km in diameter, regional devastation over an area $> 10^5 \text{ km}^2$ and, if impacting the ocean, a basin-wide tsunami. The energy is $\sim 1,800$ times that of the Chelyabinsk event.

4.3 Case B: Impact on the Moon (working hypothesis of the poster)

The impact energy on the lunar surface would be lower, but still catastrophic:

$$E_L = \frac{1}{2}(4.3 \times 10^{10})(7,800)^2 \approx 1.3 \times 10^{18} \text{ J} \approx 310 \text{ Mt of TNT} \quad (8)$$

This impact would generate a new crater of $\sim 5\text{--}8$ km in diameter (a “real-time Einstein”), an impact quake with a duration > 10 minutes—due to the absence of attenuation by fluids in the lunar rock—and a fraction of the ejecta with velocities exceeding the lunar escape velocity ($v_{esc,L} = 2.38 \text{ km/s}$) would escape into cislunar space.

4.4 Case C: Collateral damage to satellites (“it takes out several”)

(i) **On the entry path:** the 2029 flyby passes at $\sim 31,000\text{--}38,000$ km from Earth’s center, inside the geostationary ring ($r_{GEO} = 42,164$ km). Although the individual collision probability is low, a single impact at $\sim 5\text{--}7 \text{ km/s}$ with a 1 t satellite would release an energy of $E \approx \frac{1}{2}(1,000)(6,000)^2 \approx 1.8 \times 10^{10} \text{ J}$ ($\sim 4.3 \text{ t}$ of TNT), comparable to the Falcon 9 impact, generating a destructive debris cloud in the already congested GEO orbit. (ii) **By secondary ejecta after lunar impact (the central argument):** the lunar ejecta cone with $v_e > 2.38 \text{ km/s}$ becomes a rain of projectiles that would reach the distance of GEO orbit in a transit time $t \approx 3.8 \times 10^8 \text{ m} / 2.6 \times 10^3 \text{ m/s} \approx 40$ hours. This transforms a point threat into a distributed and persistent risk over all space assets in GEO and MEO, a scenario that no terrestrial catalog is designed to track today.

5 The Moon as Geological Record and Witness Plate

The surface and internal structure of the Moon offer an almost intact geological record of the first hundreds of millions of years of the solar system, functioning as a passive witness plate of impact history. Its fundamental characteristics are:

- **Crust and mantle:** The crust, 30–50 km thick, is composed of anorthosite in the highlands and Fe- and Ti-rich basalt in the maria. The peridotitic mantle extends to a small, partially liquid metallic core of 300–400 km radius.
- **Regolith:** Layer of dust and fragments produced by billions of years of bombardment. It contains oxygen trapped in silicates, ilmenite (FeTiO_3), and helium-3 from the solar wind. Ilmenite is, in addition, a potential source of ISRU raw material for manufacturing densified impact masses.
- **Magnetic fields:** Without an active global dynamo (it shut down 1.0–1.5 Ga ago), the Moon preserves remanent magnetism in the crust, notably the *Lunar Swirls* (e.g., Reiner Gamma), where intense local fields deflect the solar wind.
- **Radiation environment:** The surface, without atmosphere or global magnetosphere, receives $\sim 600 \mu\text{Sv/day}$ ($\sim 200\times$ the terrestrial surface), with a significant component of secondary neutron radiation. PSRs offer thermal stability for the observatory.
- **Lunar seismicity:** Deep quakes are distinguished (700–1,200 km, due to terrestrial tides, M 1–2), shallow quakes ($< 100 \text{ km}$, M > 5.5 , with durations $> 10 \text{ min}$ due to the absence of attenuation by fluids), thermal quakes, and **impact quakes**. An impact like that of Case B

would be seismically recorded across the entire Moon, validating its use as a calibrated detector of the impactor flux.

6 Advantage of the Lunar Platform and Kinetic Interception Architecture

6.1 Comparison of launch from Earth vs. Moon

The fundamental advantage of a lunar defense platform lies in the drastic reduction of the ΔV budget. With $v_{esc,\oplus} = 11.19$ km/s and $v_{esc,L} = 2.38$ km/s, an illustrative ΔV budget toward an interception trajectory is ~ 12.5 – 13.5 km/s from the Earth’s surface and only ~ 2.9 – 3.4 km/s from the lunar surface (a factor of ~ 4). Applying the Tsiolkovsky rocket equation with an $I_{sp} = 350$ s, the mass ratio (initial mass / final mass) drops from $e^{13,000/3,434} \approx 44$ from Earth to $e^{3,200/3,434} \approx 2.5$ from the Moon. This implies that, from the Moon, most of the launched mass is useful impact mass, and not propellant.

6.2 The “Interception Train” (IT, $N = 3$) for objects <100 m and warning <5 years

For small objects detected with little advance notice, the strategy of “deflection” gives way to that of “controlled fragmentation”. A system of three vehicles launched sequentially is proposed:

1. **Unit 1 (Pathfinder):** Performs the approach, maps the surface geology and internal porosity of the asteroid using LIDAR and spectrometers.
2. **Unit 2 (Pre-Fracture Impactor):** A high-mass kinetic impactor (tungsten or densified ISRU material) impacts at the identified weak point, creating a crater and a network of internal fractures.
3. **Unit 3 (Fragmentation Charge):** A chemical explosive charge penetrates the crater of Unit 2 and detonates, with daisy-chain telemetry from previous units to adjust the detonation in real time.

6.3 Momentum transfer and fragmentation equations

The total momentum transfer is the sum of the impulses:

$$I_{total} = \sum_{i=1}^N m_i (v_i + v_{ast}) \quad (9)$$

The structural weakening factor (σ) is reduced sequentially:

$$\sigma_{final} = \sigma_{initial} - \left(\frac{E_{cin,2}}{V_{ast}} \cdot \kappa \right), \quad E_{cin,2} = \frac{1}{2} m_2 v_{rel}^2 \quad (10)$$

The condition for safe fragmentation is that the energy of Unit 3 (E_{total}) exceeds the gravitational and cohesive binding energy of the asteroid (U):

$$U = \frac{3}{5} \frac{GM_{ast}^2}{R_{ast}} \approx \frac{3}{5} \frac{(6.674 \times 10^{-11})(4.3 \times 10^{10})^2}{157.5} \approx 4.7 \times 10^8 \text{ J} \quad (11)$$

$$E_{total} = \eta \cdot (E_{cin,3} + \Delta H_{chem}) > U + \Delta H_{fracture} \quad (12)$$

Telemetry allows Unit 3 to correct its trajectory based on the damage observed by the previous units:

$$\vec{v}_{3,adjusted} = \vec{v}_{3,nominal} + \int_{t_0}^{t_{impact}} \frac{\Delta \vec{p}_{telemetry}}{m_3} dt \quad (13)$$

Key finding: The binding energy $U \sim 10^8\text{--}10^9$ J is ~ 100 times less than the kinetic energy of a single 1 t impactor at 10 km/s (5×10^{10} J). Therefore, fragmentation is energetically trivial. The real design challenge lies in the coupling efficiency η and in controlling the size distribution of the resulting fragments, which must be < 50 m so that the Earth’s atmosphere dissipates them harmlessly, thus mitigating the “shotgun effect”.

7 Traceability to the L-RAIL Architecture (Phase C)

The MIR/IR telescope (DWG-MIR-01) is the direct response to the detection failure exposed by “Green Lantern”. Its sensitivity of $< 10 \mu\text{Jy}$ allows detecting a 140 m PHA at > 0.1 AU. The PDA-ALERT protocol transmits the alert to the TUG via the DSOC optical link (10 Gbps) with a total latency < 10 minutes, and the PDA-TARGET data format (CCSDS 734.2-B-1, DTN) is designed to feed the future interceptor silo system without requiring modifications to the observatory. The Moon, therefore, simultaneously provides the eye (MIR/IR in PSR), the witness plate (crater and seismic record), and the launcher (low v_{esc} and ilmenite ISRU).

7.1 The 2026 JH2 flyby: a “city bomb” with 8 days of advance warning

Coeval with the preparation of this mission, asteroid 2026 JH2, with an estimated diameter between 32 and 71 m (mean ~ 50 m, mass $\sim 1.7 \times 10^8$ kg), passed at 0.25 lunar distances ($\sim 96,000$ km) traveling at 20 km/s. It was discovered only 8 days before its closest approach, after emerging from the solar blind zone. In the event of impact with Earth, the released kinetic energy would have been:

$$E_{JH2} = \frac{1}{2}(1.7 \times 10^8 \text{ kg})(20,000 \text{ m/s})^2 \approx 3.4 \times 10^{16} \text{ J} \approx 8 \text{ Mt of TNT}, \quad (14)$$

equivalent to ~ 500 Hiroshima bombs or a **Tunguska**-scale event. An impact of this magnitude on a densely populated city would result in total devastation within a radius of 15–20 km. The operational lesson is critical: an 8-day advance warning is insufficient to deflect or evacuate, validating the need for continuous mid-infrared detection from platforms outside the solar blind zone, such as the proposed MIR/IR in the lunar PSR.

8 Suggested Poster Script (Pasadena)

1. **Hook:** “Green Lantern” and Chelyabinsk: detection arrives at 80–90 km altitude, without response time.
2. **Calibration:** The Falcon 9 impact on the Einstein crater is an experiment of known mass and velocity that validates π -scaling and material ejection models (5–50 km).
3. **Scenario H-2029:** Apophis crosses inside GEO; if it impacted the Moon (~ 310 Mt), the escaped ejecta would become a rain of projectiles over satellites at 36–48 hours.
4. **Solution:** The MIR/IR observatory in the PSR (alert in < 10 min) and the Interception Train launched from the Moon (ΔV four times less than from Earth).
5. **Legacy:** The Moon as witness plate and cislunar service station for the Artemis program and permanent planetary defense.

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